

Computational Physics Sommersemester 2025

Exercise Sheet 05

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1 Part a

Berendsen thermostat: We rescaled the velocities by a factor λ with the formula given for the Berendsen thermostat,

$$\lambda = \sqrt{1 + \frac{\Delta t}{\tau} \left(\frac{T_0}{T(t)} - 1 \right)}, \quad (1)$$

where Δt is the simulation step and τ a coupling parameter which determines how tightly the bath and the system are coupled together.

So each timestep we rescaled the velocities as you can see in the Code below:

```

1  def berendsen_thermostat(vx, vy, vz, T1, T2, tau, dt):
2
3      multiplier = math.sqrt(1 + (dt / tau) * ((T1 / T2) - 1))
4      vx = vx * multiplier
5      vy = vy * multiplier
6      vz = vz * multiplier
7      return vx, vy, vz

```

where T2 is $T(t)$.

We performed a temperature jump experiment with the following parameters:

N	10^3
ρ	$0.5 \sigma^{-3}$
ϵ	$0.5 k_B T_0$
τ	$5000 \Delta t$
$T_0^{(1)}, T_0^{(2)}$	300 K, 100 K

Table 1: Parameters for temperature jump experiment

In fig. 1, one can see that it takes quite some time until the temperature converges to the desired temperature $T_0^{(1)} = 300K$. This is due to the initial configuration being “unphysical” and justifies why in simulations, it is often required to conduct an equilibration run with a thermostat

before the system is in a “physical” state where the production run delivers reliable results.

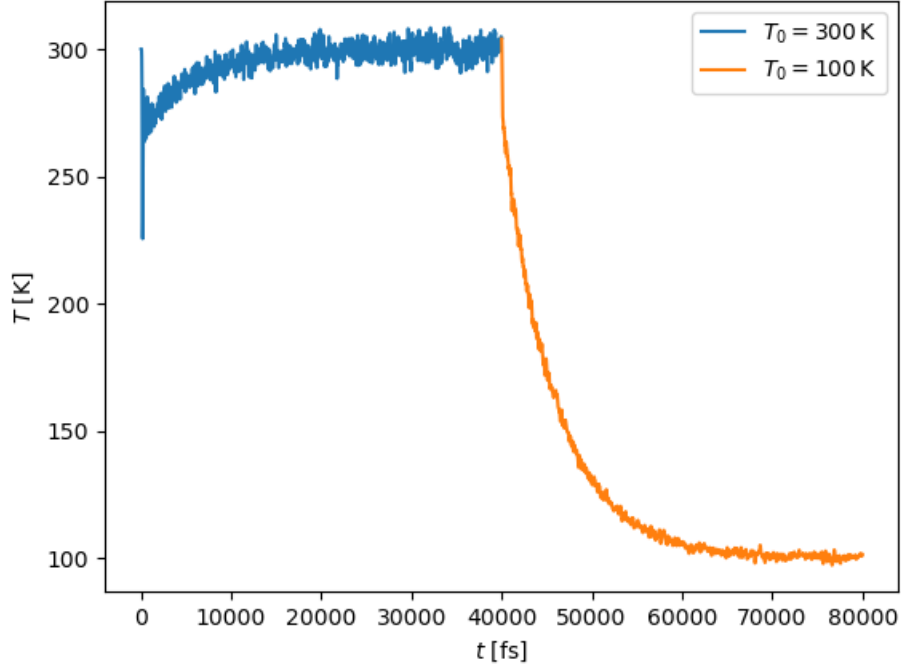


Figure 1: $T(t)$ for the Berendsen thermostat.

when applying a new temperature $T_0^{(2)} = 100K$, the system again will take some time before it decays exponentially to the desired temperature. The velocity of this decay depends on the coupling factor τ between the virtual heat bath and the system..

If the coupling factor is $\tau \rightarrow \infty$, fig. 1 would converge to 1, so there would be no rescaling, so the thermostat wouldn’t do anything at all. If the coupling factor is exactly $\tau = \Delta t$, the Berendsen thermostat describes exactly the velocity rescaling we know from the former sheets.

2 Part b

Andersen thermostat:

$$f(v) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{v^2}{2\sigma^2}\right) \quad (2)$$

As given in the exercise sheet, one can see the normal distribution in 2. By choosing a variance of

$$\sigma^2 = \frac{k_B T_0}{m} \quad (3)$$

and a mean of $\mu = 0$, this results in the Maxwell-Boltzmann distribution.

This little trick was helpful, because it allowed us to use `np.random.normal()`. In order to implement the collision rate ν : we created a random value between 0 and 1 and checked if its smaller than the probability of collision $\nu * \Delta t$. In order to get the random value we used `numpy.random.rand()` as can be seen in the python implementation below.

```
1 def anderson_thermostat(vx, vy, vz, nu, T0, dt):
2     convunits = 238845.9 * 3/2
3     variance = np.sqrt(settings.kb * T0 / settings.mass / convunits)
4     for i, _ in enumerate(vx):
5         if np.random.rand() < nu * dt:
6             vx[i] = np.random.normal(0, variance)
7             vy[i] = np.random.normal(0, variance)
8             vz[i] = np.random.normal(0, variance)
9     return vx, vy, vz
```

We performed a temperature jump experiment again and obtained the behavior visualized in fig. 2.

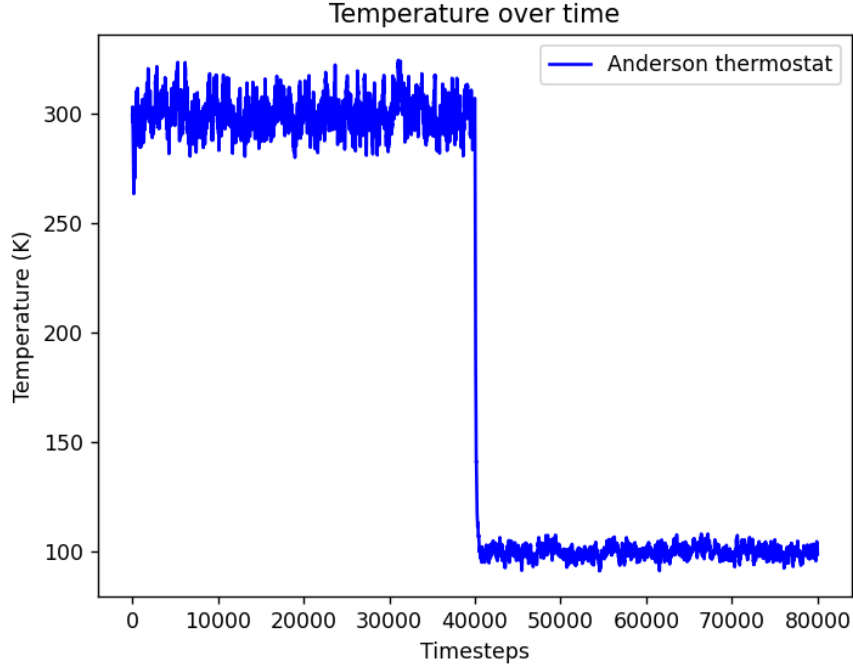


Figure 2: Temperatur jump calculation with Andersen thermostat and a collision rate of $\nu = 0.01 fs^{-1}$. The initial temperatur T_0 was 300K and after the jump it was 100K.

Compared to Berendsen thermostat visualized in fig. 1, the temperature adapts very fast to the temperature, which makes sense since the system “forgets” its former states due to the random attribution of velocities, so there are no correlations to former timesteps after a Andersen collision.

If the collision rate ν converges to zero, we dont see any ‘rescaling’ as this is equivalent to deactivating the thermostat. For $\nu = 1$ and thus more collisions, the fluctuations become much smaller, however one can discuss whether this describes a “real” MD-simulation since the particles can’t propagate freely between the rescaling steps.

3 Part c

Using the parameters displayed in table 2, we made 40000 timesteps in an equilibration run and used 40000 timesteps in a production run. The Berendsen thermostat was activated the entire time.

N	10^3
ρ	$0.5 \sigma^{-3}$
ϵ	$0.5 k_B T_0$
τ	$500 \Delta t$
T_0	300 K

Table 2: Parameters for part c.

From the production run, we were able to calculate the values $\langle T \rangle$, $\langle E \rangle$ and $\langle \sigma_E^2 \rangle$ (and even some more which we considered meaningful...). These can be found in section 3, where we also display values of the further tasks for a better comparison later on.

Property	Berendsen Thermostat	Andersen Thermostat
$\langle T \rangle$ [K]	299.97	300.24
$\langle \sigma_T^2 \rangle$ [K ²]	9.64	59.11
$\langle E \rangle$ [kcal/mole]	322.94	324.66
$\langle \sigma_E^2 \rangle$ [(kcal/mole) ²]	18.88	903.70
$\langle K \rangle$ [kcal/mole]	1043.72	1044.68
$\langle \sigma_K^2 \rangle$ [(kcal/mole) ²]	116.77	715.67
$\langle U \rangle$ [kcal/mole]	-720.78	-720.02
$\langle \sigma_U^2 \rangle$ [(kcal/mole) ²]	169.22	188.26
$\langle U \rangle$ [kcal/mole] via $g(r)$	-16319.67	-16309.05

Table 3: Values from part c, d and e

Specifically for this task, section 3 shows that the mean temperature is indeed very close to 300 K. The comparison between $\langle E \rangle$ and $\langle \sigma_E^2 \rangle$ shows that the fluctuations of the total energy are quite small, which makes also sense since the Berendsen thermostat can be interpreted as a mix between a NVE and a NVT ensemble.

4 Part d

For the calculation of the mean potential energy $\langle U \rangle$ via the $g(r)$, we used eq. (4) which was given on sheet #3:

$$U = N \cdot \frac{\rho}{2} \int d\mathbf{r} V_{\text{LJ}}(r) g(r) \quad (4)$$

We computed the resulting $g(r)$ as well as the integral core (converged to spherical coordinates, as successfully done in the last sheet), which is displayed in fig. 3.

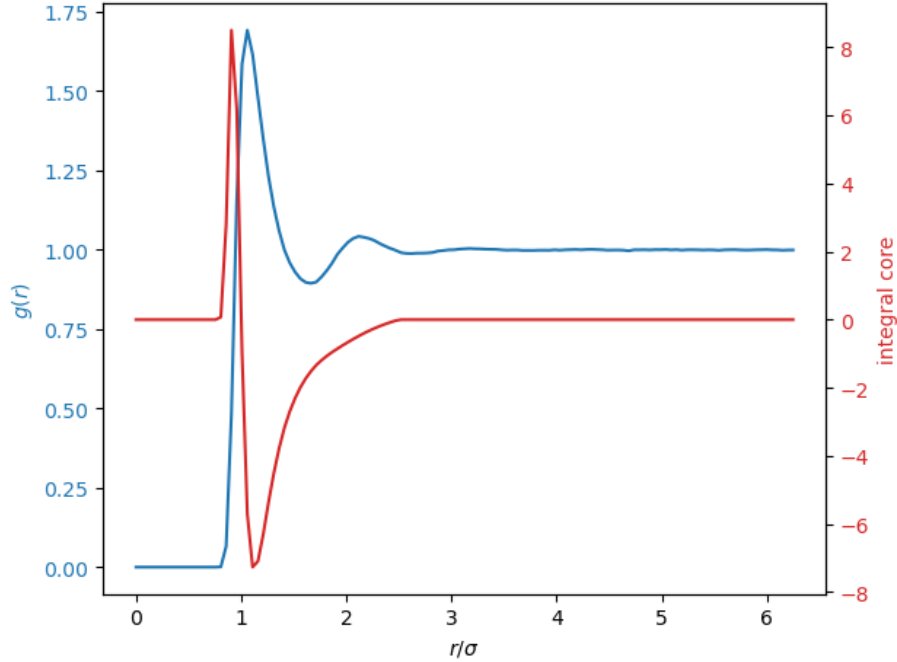


Figure 3: $g(r)$ and the integral core using the Berendsen thermostat.

With the integral core, we now computed $\langle U \rangle$ using eq. (4), which delivered us a value of -16319.67 kcal/mole, which is way too large compared to the mean potential energy obtained by using the fact that the system should be ergodic and thus averaging over $U(t)$ as done in part c. For a better comparison, these values can all be found in section 3.

If we take a look at the $g(r)$ and the integral core in fig. 3, we see exactly the expected behavior, so we're sure that this huge deviation of the values only stems from a wrong conversion factor. This argument will be solidified even more later.

5 Part e

In task e, we repeated the same steps as in c and d, but now with the Andersen thermostat instead of the Berendsen thermostat. All values can be found in section 3.

Like in part d, the $g(r)$ and the integral core displayed in fig. 4 look like expected and almost the same as for the Berendsen thermostat in fig. 3 (and before you wonder: they are not the same picture, they just look very similar).

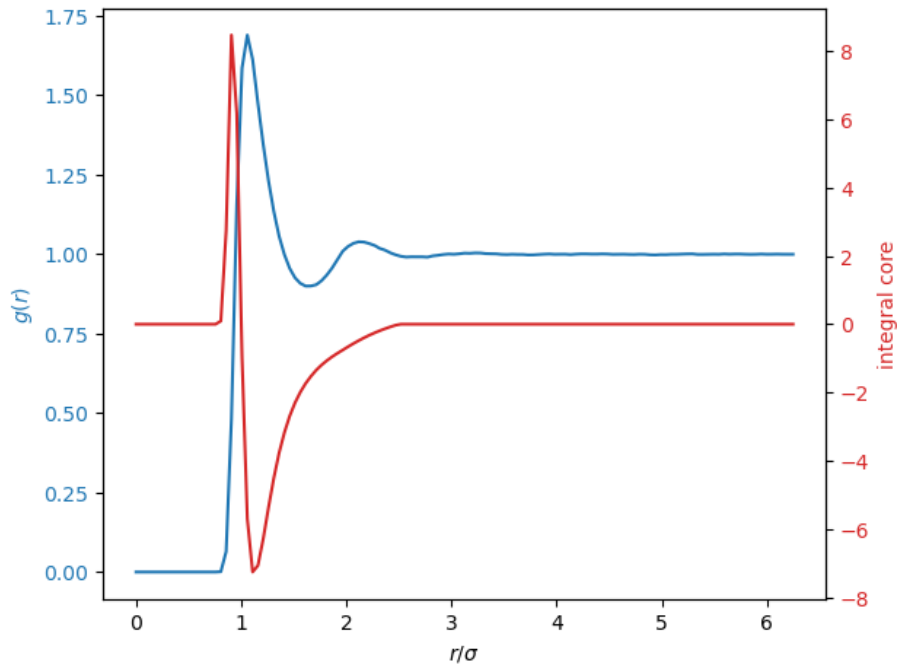


Figure 4: $g(r)$ and the integral core using the Andersen thermostat.

It is thus no surprise that the calculated value of $\langle U \rangle$ using

eq. (4) is very similar as for the Andersen thermostat. This solidifies the suspicion that it just a prefactor which is wrong, but after our experience with sheet 1, we decided to cancel the useless search for this prefactor after half an hour since we don't learn anything from it...

What we learned from the values of the Andersen thermostat, however, is that the mean temperature is also very close to the desired 300 K. Also the mean values for the total, potential and kinetic energy are very similar to what we saw in the Berendsen thermostat. However, we can clearly see that for the Andersen thermostat, the variance and thus the fluctuations are much stronger. After some thinking, this is very reasonable since, as we said earlier, the Berendsen thermostat doesn't model a "proper" NVT ensemble, but rather a mix with an NVE ensemble. Therefore, the energy fluctuations in the Berendsen thermostat are smaller than for the Andersen thermostat, which is a "proper" NVT thermostat and thus allows more fluctuations of the energy.